

“SO LONG AS THE MOTHER, IGNORANCE, LIVES, IT IS NOT SAFE FOR SCIENCE, THE OFFSPRING, TO DIVULGE THE HIDDEN CAUSE OF THINGS.”

Johannes Kepler, German astronomer

ON OCTOBER 2, 1608, HANS LIPPERSHEY (1570–1619), a lens-maker in Zeeland in the Dutch Netherlands, applied for a patent for a device for “seeing things far away as if they were nearby.” This was soon known as a **telescope**. Lippershey’s device was crude, only magnifying by three times, and was soon exceeded by others. But it was still a milestone in the development of scientific observation in 17th-century Europe.

Since 1606, the Dutch had been trying to broker a truce with Spain to halt the ongoing wars of the **Dutch Revolt**. Forty years of war had left both sides spent, yet each feared the other would use a ceasefire to regroup—as each intended to do. Despite this, in April 1609, a **12-year truce** was agreed.

In 1526, Charles V had decreed that all Muslims in Spain convert to Catholicism. The resulting

minority **Morisco** population remained on the margins of Spanish society—valued for their cheap labor, but suspected for their religious affiliation. In 1609, Philip III agreed to **expel** them from Spain entirely. The decision caused whole communities to be summarily expelled and their possessions forfeited. It also created economic dislocation in many parts of the country as a valuable source of labor disappeared. Muslim resentment toward Spain predictably increased.

In 1609, the Dutch East India Company had sent **Henry Hudson** to investigate North America’s east coast. He explored the Hudson River to present-day Albany, claiming the region for the Dutch.

Hudson River

New York State’s river is named for Englishman Henry Hudson, who explored the river’s course.

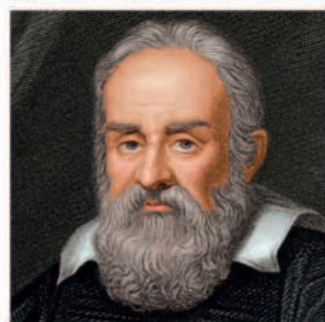


A copper engraving depicts the assassination of Henry IV, King of France, in Paris. Henry IV had survived 18 previous attempts on his life.

THE ASSASSINATION OF HENRY IV

in Paris on May 14, 1610, stabbed by a one-time monk and teacher, Francois Ravaillac, promised a reawakening of the brutal religious divisions Henry had worked so hard—and killed so many—to avoid. The reality was quite the opposite. Not only was Henry’s nine-year-old son immediately accepted as the new monarch, **Louis XIII**, but the threat of renewed conflict between France and Spain was averted. Both had been sparring for control of the duchy of Jülich-Cleves in Germany, threatening a renewed pan-European religious conflict. With Henry’s death, both could now legitimately retire with no loss of face. Henry IV, first of the Bourbon kings, was among the most remarkable of France’s kings: his **reconstruction** of the pestilential medieval shambles of Paris echoed his far-sighted reconstruction of France itself.

Less than a year after Hans Lippershey claimed to have invented a telescope, **Galileo** (see panel, right), working from no more than descriptions of Lippershey’s device, had devised his own. It took him, he claimed, less than one day to put together. It was with this basic instrument that, in January 1610, Galileo began to observe the “three fixed stars,” invisible to the naked eye, that were next to the planet Jupiter. They were, he realized, **orbiting** the planet. This was a discovery that challenged the accepted notion of how heavenly



GALILEO (1564–1642)

Galileo Galilei, born in Pisa, was an Italian scientist who, despite obstruction from religious orthodoxy, revealed an entirely new, scientific understanding of the world. The Church regarded his revelations as heresy but, reluctant to condemn the scientific pioneer outright, did its best to accommodate him. Heretic or not, Galileo died with his reputation not just growing but assured.

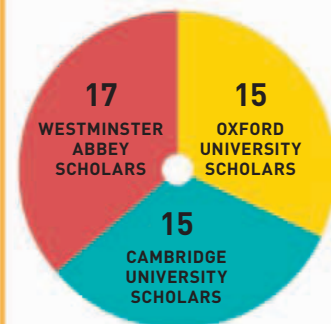
bodies could orbit only one fixed point in the skies: the Earth. This explosive revelation was reinforced later the same year when Galileo began a systematic series of observations of the planet Venus. Its phases—crescent, partial, and full—could be explained only if it, too, was orbiting another body, the **Sun**. Observations made possible by the telescope were poised to revolutionize humanity’s understanding of its relationship with a vast, impersonal universe.



This painting shows the Battle of Kalmar on the Baltic Sea.

IN 1604, KING JAMES I OF

ENGLAND authorized a new English translation of the **Bible**. Since the Reformation there had been two previous English translations: the *Great Bible* of 1539 and the *Bishops’ Bible* of 1568. However, it was felt that both contained minor **inaccuracies** and neither fully reflected the doctrinal authority and structure of the **Church of England**. The new translation, published in 1611 as *The Holy Bible*, was the work of 47 scholars under the direction of the Archbishop of Canterbury, Richard Bancroft. Though accepted relatively slowly by the Anglicans, by the 18th century it was widely regarded by all English-speaking Protestant churches as the **definitive** English-language Bible. It was only when the revised edition was issued in the late 18th century that it became commonly referred



King James Bible

Several scholars from each institution translated the Bible from Greek, Hebrew, and Latin into English in 1604–08.

1608 Spain legalizes slavery of Chilean Indians

1608 French explorer Samuel de Champlain (c. 1567–1635) founds Quebec

1608 Work restarted on key Roman High Baroque church of Sant’Andrea della Valle, Rome (completed 1650)

1608 First telescope made by Hans Lippershey, Netherlands

April 9, 1609 Twelve Years’ Truce agreed between Spain and Netherlands

1609 Dutch expedition under Henry Hudson explores Hudson River

1609 Moriscos expelled from Spain

1609 Astronomia Nova by Johannes Kepler (1571–1630) published; outlines first two laws of planetary motion

1609 Samuel de Champlain explores St. Lawrence and eastern Great Lakes, North America

1609 Polish war with Russia (to 1618)

January Galileo Galilei reports his discovery, via use of the telescope, of moons of Jupiter

May 14 Assassination of Henry IV of France (b. 1553)

August Hudson Bay, North America, discovered by Henry Hudson

Dahomey kingdom established, West Africa

Oromo are dominant population in South Ethiopia and prominent among Muslims of the Harar plateau

Poles occupy Moscow after Swedish invasion of Russia

English trading post established at Masulipatam, East India